# ESE 123 Lab 5 Report: Capacitors and Oscilloscope

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**Objective.** Practice the DSO-X 3012A's cursors and automatic measurements, measure the frequency response of an RC high-pass filter, and check the voltage divider on the project board.

## 1. Scope measurements (1 $V_{rms}$, 900 Hz sine, probe at node "top")

| | Counting boxes | Cursors | Automatic |
|---|---|---|---|
| $V_{pp}$ | 5.6 V | 5.67 V | 5.79 V |
| Period | 1.11 ms | 1.114 ms | 1.111 ms |
| Frequency | ≈ 900 Hz | 897.6 Hz | 900 Hz |

AC RMS (N cycles), channel 1: **1.9827 V**. This is about twice the 1 $V_{rms}$ setting because the generator assumes a 50 Ω load, but the circuit is a high-impedance load. Expected $V_{pp} = 2\sqrt{2} \times 1.98 \approx 5.61$ V. All three methods agree to within about 3%. The automatic measurement is the most convenient, and the box count is the least precise.

## 2. Frequency response of the RC high-pass filter (R2 = 4.75 kΩ, C1 = 100 nF)

$f_c = 1 / (2\pi \cdot R \cdot C) = 1 / (2\pi \cdot 4750\ \Omega \cdot 100\ \text{nF}) \approx 335$ Hz

$V_{mid}\ (\text{theory}) = V_{top} \cdot (f/f_c) / \sqrt{(1 + (f/f_c)^2)}, \quad V_{top} = 1.98\ V_{rms}$

| Frequency | $V_{mid}$ measured | $V_{mid}$ theory | Phase measured | Phase theory (lead) |
|---|---|---|---|---|
| 3 kHz | — | 1.97 V | −4.2° | 6.4° |
| 1 kHz | 1.856 V | 1.88 V | −16.5° | 18.5° |
| 900 Hz | 1.8249 V | 1.86 V | −20° | 20.4° |
| 300 Hz | 1.29 V | 1.32 V | −45° | 48.2° |
| 100 Hz | 568 mV | 567 mV | −67° | 73.4° |
| 30 Hz | 188 mV | 177 mV | −64° | 84.9° |
| 10 Hz | 85 mV | 59 mV | −43° | 88.3° |
| 3 Hz, 1 Hz | No usable signal: too small and slow for the scope to measure | | | |

The scope reports a negative phase, which means channel 2 ($V_{mid}$) leads channel 1. The plots use the magnitude of the phase. Dots are measured values and the dashed line is theory.

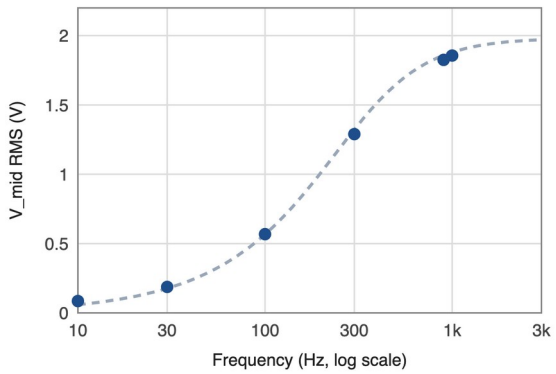



*Figure 1: RMS voltage vs. frequency*

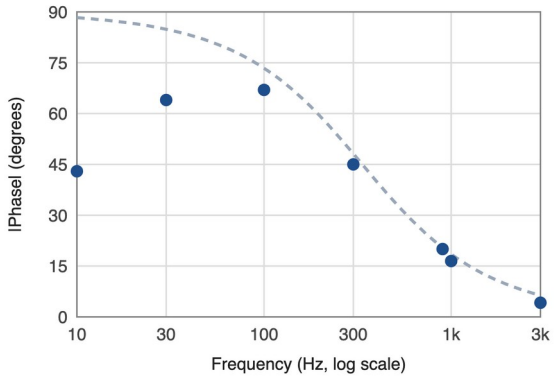



*Figure 2: Phase vs. frequency*

**Discussion.** The circuit behaves as a high-pass filter. Above about 1 kHz almost all of the input reaches $V_{mid}$, and below the 335 Hz corner the output falls off steadily. The magnitude matches theory within a few percent from 100 Hz to 1 kHz. At 30 Hz and 10 Hz the signal is only tens of millivolts, so noise and the scope's resolution push the readings above theory. The phase follows the expected trend (about 45° near $f_c$, small at high frequency), but at 30 Hz and 10 Hz it drops back down instead of approaching 90°. At those levels the scope's phase measurement is no longer reliable.

## 3. Voltage divider (R6 = 59 kΩ top, R7 = 150 kΩ bottom)

| Quantity | Value |
|---|---|
| AVDD measured (DMM) | 5.079 V |
| TP2 measured (DMM) | 3.63 V |

> a) TP2 expected = AVDD · R7/(R6 + R7) = 5.079 · 150k/209k = 3.645 V
> b) % error = (3.63 − 3.645)/3.645 × 100 = −0.42 %
> c) AVDD at TP2 = 4.3 V:  AVDD = 4.3 · 209k/150k = 5.99 V

The measured TP2 is within 0.5% of the calculated value, which is consistent with 1% resistors. The microcontroller can measure supply voltages up to about 6.0 V before TP2 exceeds its 4.3 V reference.